

MSTAR PRACTICE WORKSHEET**Lesson 5-2: Determine the Area of Triangles**

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

What is the area of a triangle with base 10 cm and height 6 cm?

- ☐ A 16 sq cm
- ☒ B 30 sq cm
- ☐ C 30 cm
- ☐ D 60 sq cm

$$A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 6 = 30 \text{ square centimeters.}$$
Question 2 **SELECTED RESPONSE — CORRECT: B**

A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

- ☐ A 3 ft
- ☒ B 6 ft
- ☐ C 12 ft
- ☐ D 16 ft

$$A = \frac{1}{2} \times b \times h \rightarrow 24 = \frac{1}{2} \times 8 \times h \rightarrow 24 = 4h \rightarrow h = 6 \text{ ft.}$$

Question 3 **SELECTED RESPONSE — CORRECT: A**

Why do we divide by 2 when finding the area of a triangle?

- ☒ A A triangle is exactly half of a rectangle with the same base and height
- ☐ B Triangles have 2 equal sides
- ☐ C The base is always twice the height
- ☐ D We always divide area formulas by 2

A triangle with base b and height h fits inside a rectangle with the same base and height. The triangle covers exactly half the rectangle, so $A = \frac{1}{2} \times b \times h$.

Question 4 **SELECTED RESPONSE — CORRECT: B**

What is the area of a triangle with base 14 ft and height 6 ft?

- ☐ A 20 sq ft
- ☒ B 42 sq ft
- ☐ C 42 ft
- ☐ D 84 sq ft

$A = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42$ sq ft.

Question 5 **SELECTED RESPONSE — CORRECT: B**

A triangular window in the firm's lobby has a base of 10 ft and a height of 6 ft. What is its area?

- ☐ A 16 sq ft
- ☒ B 30 sq ft
- ☐ C 32 sq ft
- ☐ D 60 sq ft

$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 10 \times 6 = \frac{1}{2} \times 60 = 30$ sq ft. The triangle is half of the $10 \times 6 = 60$ rectangle it fits inside.

Question 6

SELECTED RESPONSE — CORRECT: B

A triangular roof panel on the blueprint has a base of 7 ft and a height of 4 ft. What is its area?

- ☐ A 11 sq ft
- ☒ B 14 sq ft
- ☐ C 22 sq ft
- ☐ D 28 sq ft

$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \times 4 = \frac{1}{2} \times 28 = 14$ sq ft. Multiply base $\times$ height first (28), then take half (14).

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

A triangular garden bed has a base of 14 meters and a perpendicular height of 6 meters. What is its area?

- ☐ A 10 square meters
- ☐ B 20 square meters
- ☒ C 42 square meters
- ☐ D 84 square meters

$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 14 \times 6 = 42$ square meters.

- **A:** That halves the sum of 14 and 6. Multiply first: half of 14 times 6.
- **B:** That adds 14 and 6. Multiply the two measurements, then halve.
- **D:** That is base times height with no halving, which gives the PARALLELOGRAM area. A triangle is half of it.

Part B — correct answer: A

Which statement explains why the area formula for a triangle includes the factor $\frac{1}{2}$?

- ☒ A A triangle is exactly half of a parallelogram with the same base and height.
- ☐ B Triangles always have three sides, so you divide by three and round.
- ☐ C The height of a triangle is always half of its base.
- ☐ D Half of any shape's area is its perimeter.

Two identical triangles fit together to form a parallelogram with the same base and height, so one triangle is half of $b \cdot h$.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: The Missing One-Half

Dev found the area of a triangle with a base of 10 cm and a height of 7 cm. He wrote: 'Area = $10 \times 7 = 70$ square centimeters.'

Explain Dev's misconception and give the correct area.

Model answer: Area = $\frac{1}{2} \times 10 \times 7 = 35$ square centimeters. Dev used $b \cdot h$, which is the area of a parallelogram; a triangle is half of that parallelogram.

2 points	Student identifies that Dev used the parallelogram formula and omitted the factor of $\frac{1}{2}$, and gives 35 square centimeters.
1 point	Student gives 35 square centimeters but does not explain why the $\frac{1}{2}$ is needed.
0 points	Student agrees with Dev or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A rectangle is 12 ft by 8 ft. A diagonal line cuts it into two triangles. What is the area of each triangle? Explain how the triangle area formula relates to the rectangle area formula.

Model answer: The rectangle's area is $12 \times 8 = 96$ sq ft, and one diagonal cuts it into two identical triangles, so each triangle is $96 \div 2 = 48$ sq ft. The formula says the same thing: each triangle keeps the rectangle's base of 12 and height of 8, and $A = \frac{1}{2} \times 12 \times 8 = 48$ sq ft. The $\frac{1}{2}$ in the triangle formula IS that diagonal cut.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.